

How simple a (simulated) critter and how few critters for emergent behavior to arise?

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Humuhumunukunukuapua`a Works Special Projects ($-2 \leq \text{TRL} \leq 0.5$)

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- Emergent behavior:
 - Background.
 - Motivation.
- Modeling approach.
- Next steps.
- Questions/Hypotheses.
- Conclusions – not yet.

What is Emergent Behavior?

Gerald E. Marsh¹:

- *“There is not even a universally acknowledged definition of emergence.”*
- *“...it is more of a collection of ideas that have in common the notion that **complex behavior can arise from the underlying simple rules of interaction.**”*

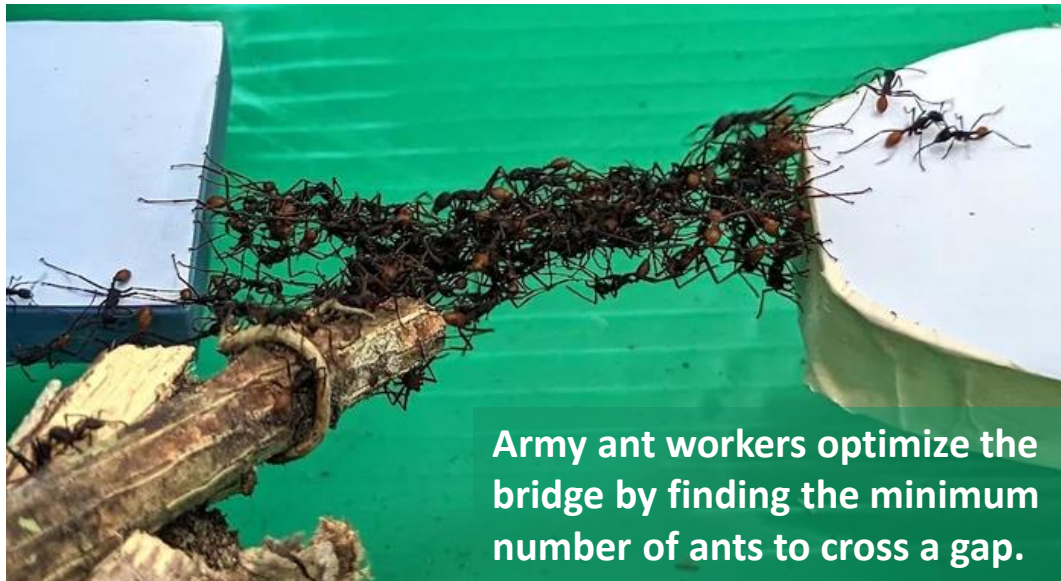
¹Marsh, Gerald E. *The Demystification of Emergent Behavior*. (arXiv General Physics, 2013).

Emergent Behavior – Examples

Ant colonies employ a simple set of rules between:

- Individual ants (social).
- Ant and environment.

Social learning occurs during the lifetime of a colony as it adapts to the environment.



“Army ants use collective intelligence to build bridges. Robots could learn from them.”²

Some believe that *“consciousness and intelligence appear to emerge gradually as the complexity of life increases.”¹*

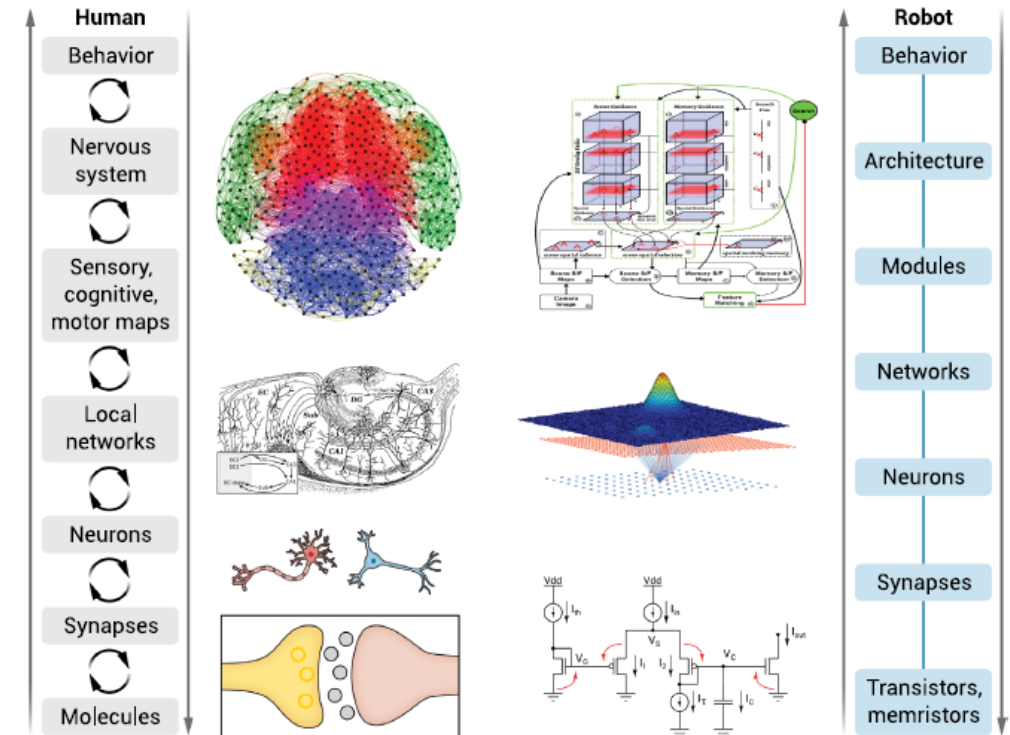


Fig. 2. Principles of information processing in biological networks and in neuromorphic circuits. The signal flow in biological and neuromorphic circuits unfolds on different spatial and temporal scales. A hierarchy of closed-loop control flows can be built both for biological neural systems and neuromorphic systems. In biology, these loops start with molecular dynamics of neurotransmitters, followed by dynamics of neurons and synapses, then local network circuits (e.g., the hippocampal circuits shown here), different sensory-motor cortical maps, ultimately leading to behavior. In a similar fashion, starting with dynamics of individual transistors and simple circuits, dynamics of neurons and synapses can be built, followed by networks, network modules, and whole architectures that can control autonomous behavior of a robot. We need to build theory, computing framework, and algorithmic understanding on these different levels of abstraction.

Reproduced from Sandamirskaya et al.³

²<https://www.npr.org/2023/11/21/1214246291/army-ants-architecture-science-robots-research> last access 21 June 2024

³From Sandamirskaya Y, Kaboli M, Conradt J, Celikel T. Neuromorphic computing hardware and neural architectures for robotics. Sci Robot. 2022;7(67)

Why Investigate Emergent Behavior?

"What I cannot create, I do not understand."

Richard P. Feynman

- Accomplish complex missions with (relatively) **simple** agents operated by on-board **trusted autonomous** algorithms capable of **on-line learning** and mathematically proven **stable**.

- Challenges:

- **Causality dilemma:**

- How do we efficiently raise the TRL of a swarm?

$$\frac{\sum_{i=1}^n \text{TRL } 9_i}{n} \neq \text{TRL } 9_{\text{swarm}}$$

- **Dissonant relationship** with current systems and operations practice:

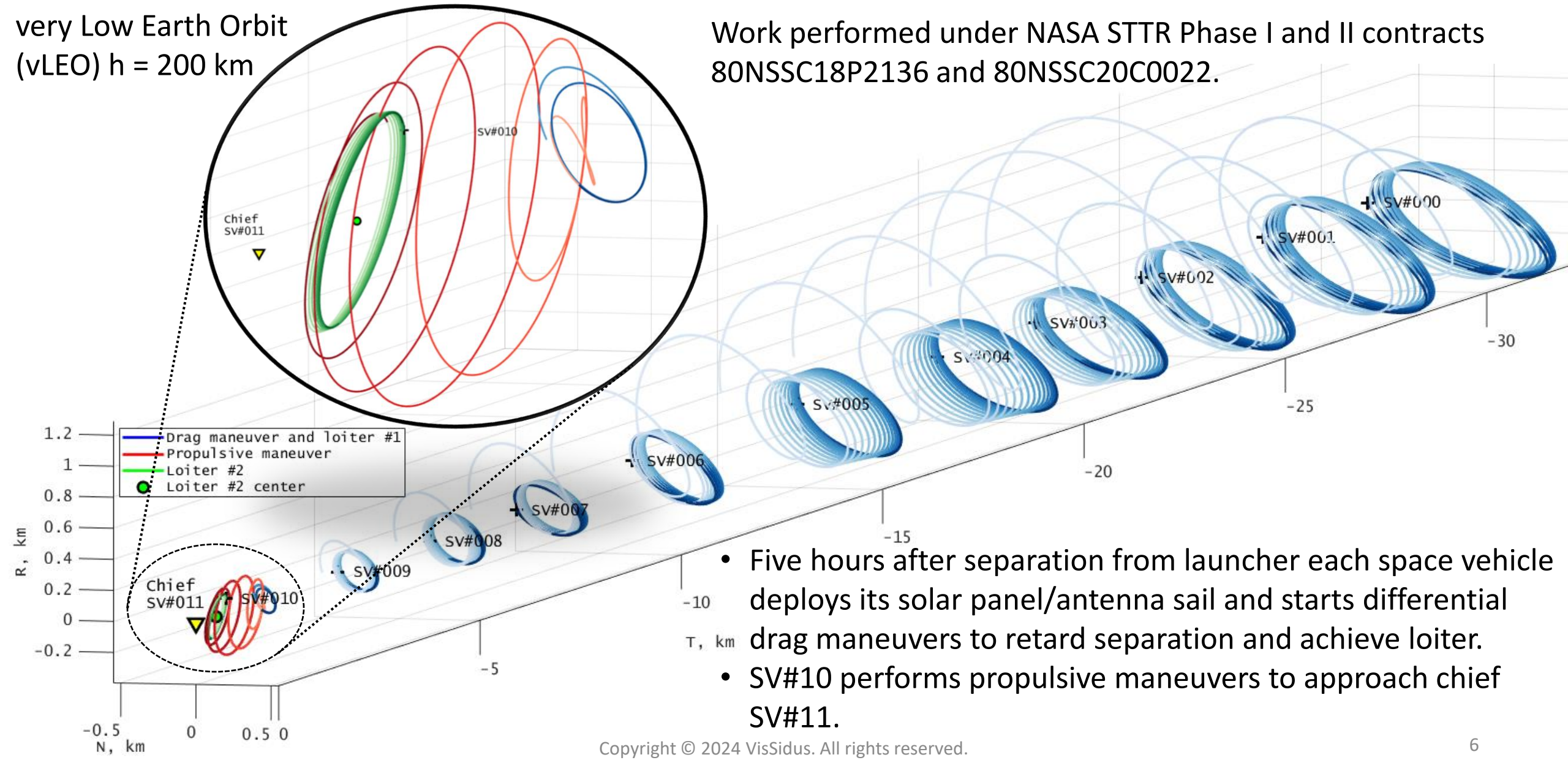
- Single space vehicle: resources and margins are allocated early in the project and mollycoddled throughout the mission life.
 - Swarm of space vehicles: how do we allocate propellant in a manner that matches current practice?
 - For example: how would systems engineers and managers handle emergent behavior that leads to propellant optimal swarm maneuvers?



Context: Spacecraft Swarm 12 L-Band SAR

very Low Earth Orbit
(vLEO) $h = 200$ km

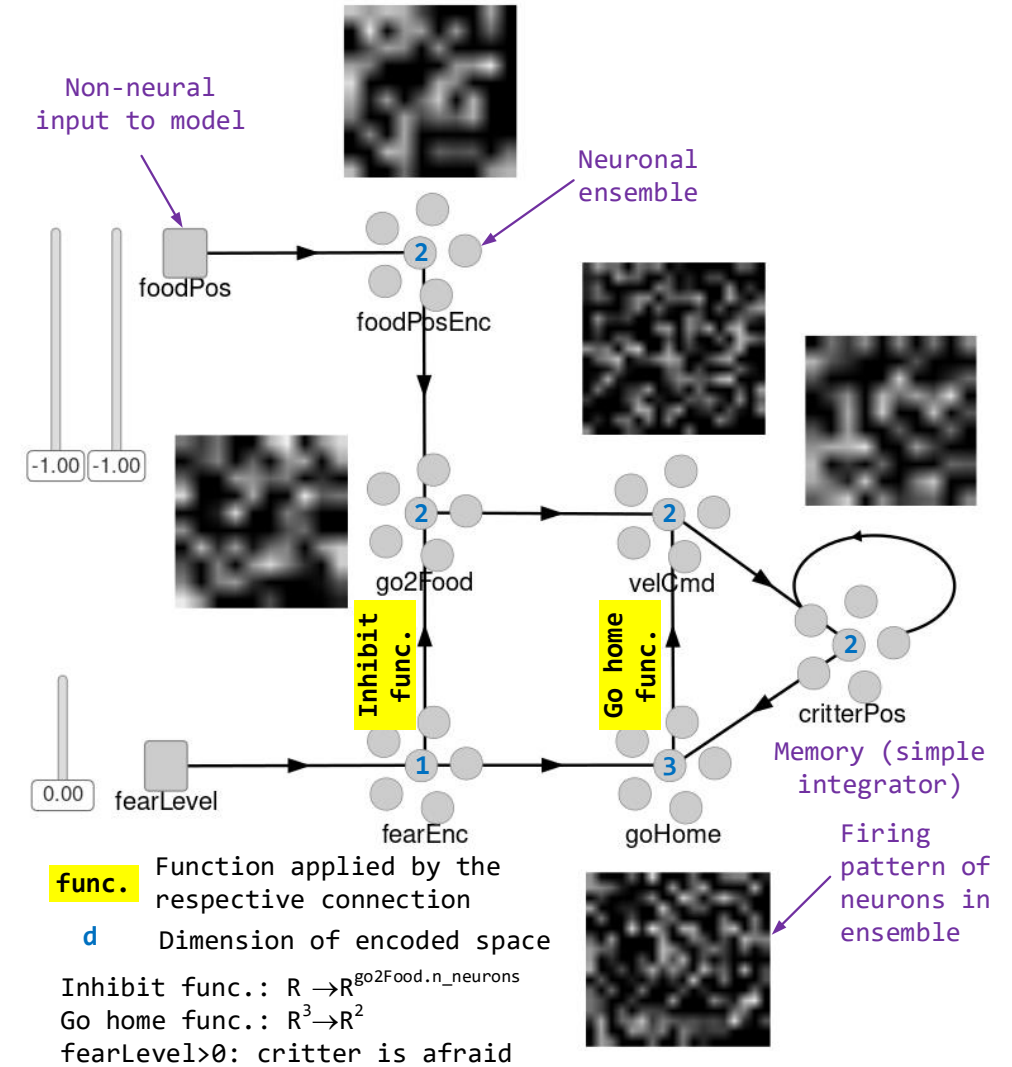
Work performed under NASA STTR Phase I and II contracts
80NSSC18P2136 and 80NSSC20C0022.



- Five hours after separation from launcher each space vehicle deploys its solar panel/antenna sail and starts differential drag maneuvers to retard separation and achieve loiter.
- SV#10 performs propulsive maneuvers to approach chief SV#11.

Approach

- Employ biologically inspired critter simulations:
 - **Neural engineering object**⁴ (Nengo) from Applied Brain Research.
- Allow critters to interact.
 - Observe emergent behavior?
 - Mathematically prove convergence to a certain behavior, i.e., steady state of the system.
- Gamify, e.g., in MicroRTS⁵:
 - Critters controlled by Nengo.
 - Critters controlled by humans.



Modified simple critter – original by Terry Stewart and Chris Eliasmith Nengo Summer School (2023)

⁴<https://www.nengo.ai/> last access 21 June 2024.

⁵Small implementation of a Real Time Strategy (RTS) game, designed to perform AI research. <https://github.com/rubensolv/MicroRTS>

Next Steps (1/2)

- Single critter of increased complexity
 - Diversify the fear model:
 - Current: “fear of predators.”
 - Include “fear” of colliding with objects, including other critters.
 - Implement a hunger model:
 - Hunger increases linearly with time.
 - Allow hunger to inhibit fear.
 - Implement a mapping skill:
 - Spatial semantic pointers.
 - Implement action selection:
 - Semantic pointers?

Next Steps (2/2)

- Multiple critters:
 - Implement critter-to-critter interaction:
 - Simple interactions: e.g., “I see you and I know my distance to you.”
 - Communication channels of varying widths.
- Backgrounder: the math.
 - Investigate contraction theory for single critter and emergent behavior of interacting critters.
 - Are there any other ways to mathematically prove stability and eventual convergence?

Reproduced from Bullo et al.⁶

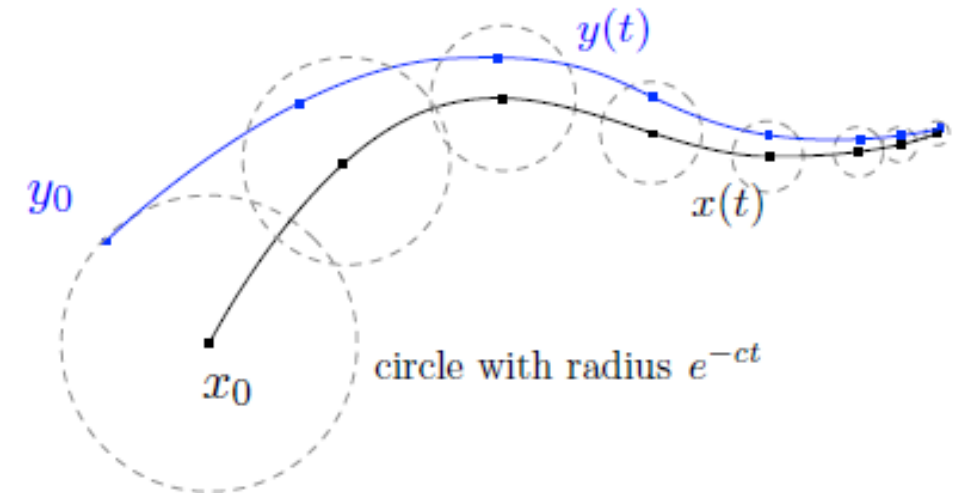


Fig. 1: Exponential incremental stability of contracting vector fields. The distance between two trajectories decreases exponentially fast.

⁶Bullo, F., Cisneros-Velarde, P., Davydov, A., & Jafarpour, S. (2022). From Contraction Theory to Fixed Point Algorithms on Riemannian and Non-Euclidean Spaces. 2021 60th IEEE Conference on Decision and Control (CDC),

How do we
**recognize emergent
behavior** in
simulations?

- Challenge: *“While the concept of emergent behavior may be difficult to define, it is easy to intuitively understand.”¹*
- Answer #1: *“Swarm motion can be interpreted as a mobile variation of Hebbian learning, given a natural translation between spatial relationships in a swarm and connectivity relationships in a neuronal network.”⁷*

How do we **quantify
and model** critter-
and group-level
decision making?

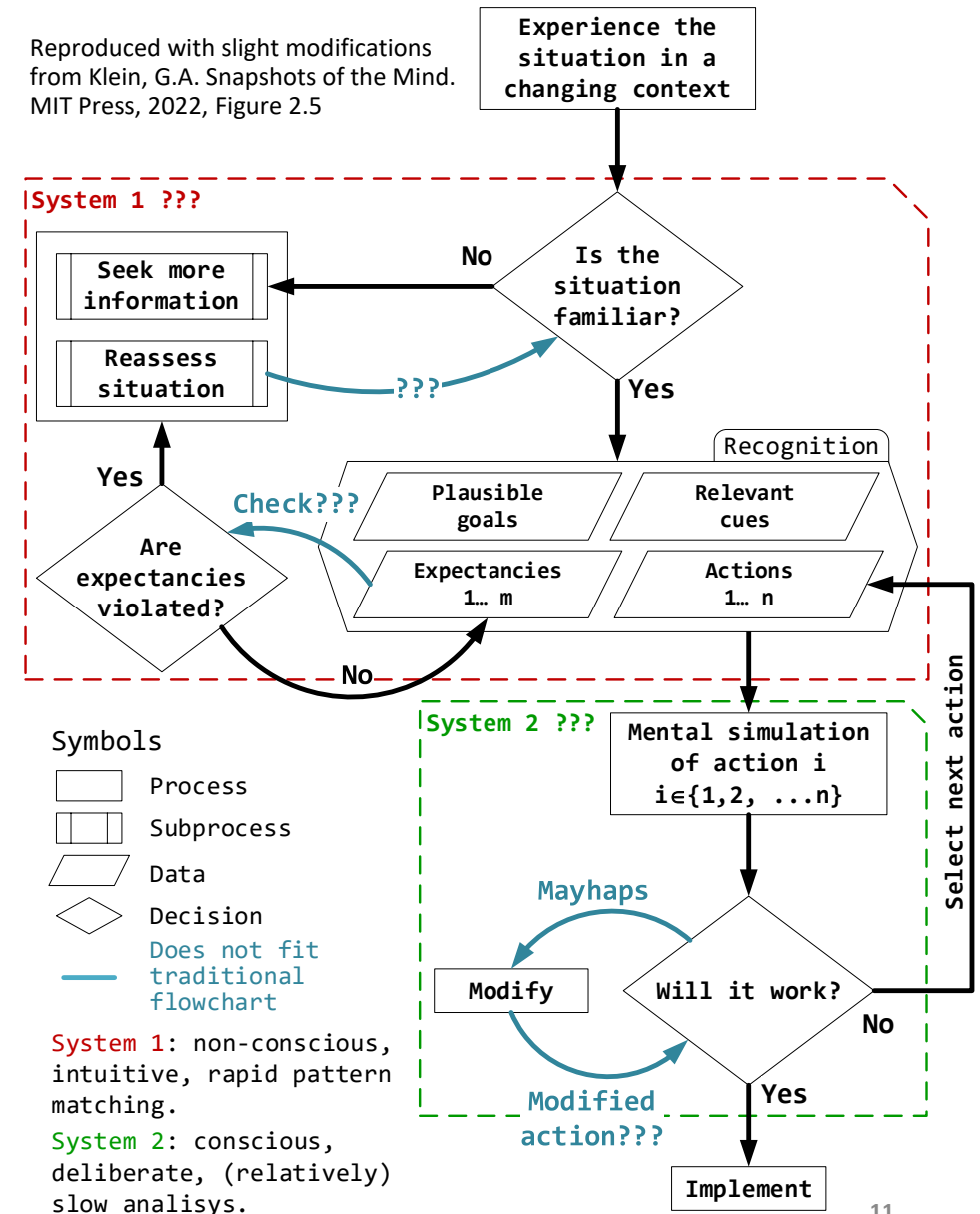
- Challenge: Identify boundaries between critter- and group-level decision making.
- Answer #1: Start with the Recognition-Primed Decision (RPD) model⁸.

⁷Hwang, Grace M., Kevin M. Schultz, Joseph D. Monaco, and Kechen Zhang. "Neuro-Inspired Dynamic Replanning in Swarms- Theoretical Neuroscience Extends Swarming in Complex Environments." Johns Hopkins APL Technical Digest 35, no. 4 (2021): 443-47.

⁸Klein, G.A. Snapshots of the Mind. MIT Press, 2022.

Backgrounder: Recognition-Primed Decision Model

- Logistics:
 - Have I drawn the System 1 and System 2 boundaries correctly?
 - If no, where should the boundaries be?
- Emerging behavior:
 - Are there any studies or results that have segregated System 1 and System 2 for simple critters?
 - Would a simple critter operate mostly in System 1 and the group/hive of simple critter operates mostly in System 2?



- None yet.

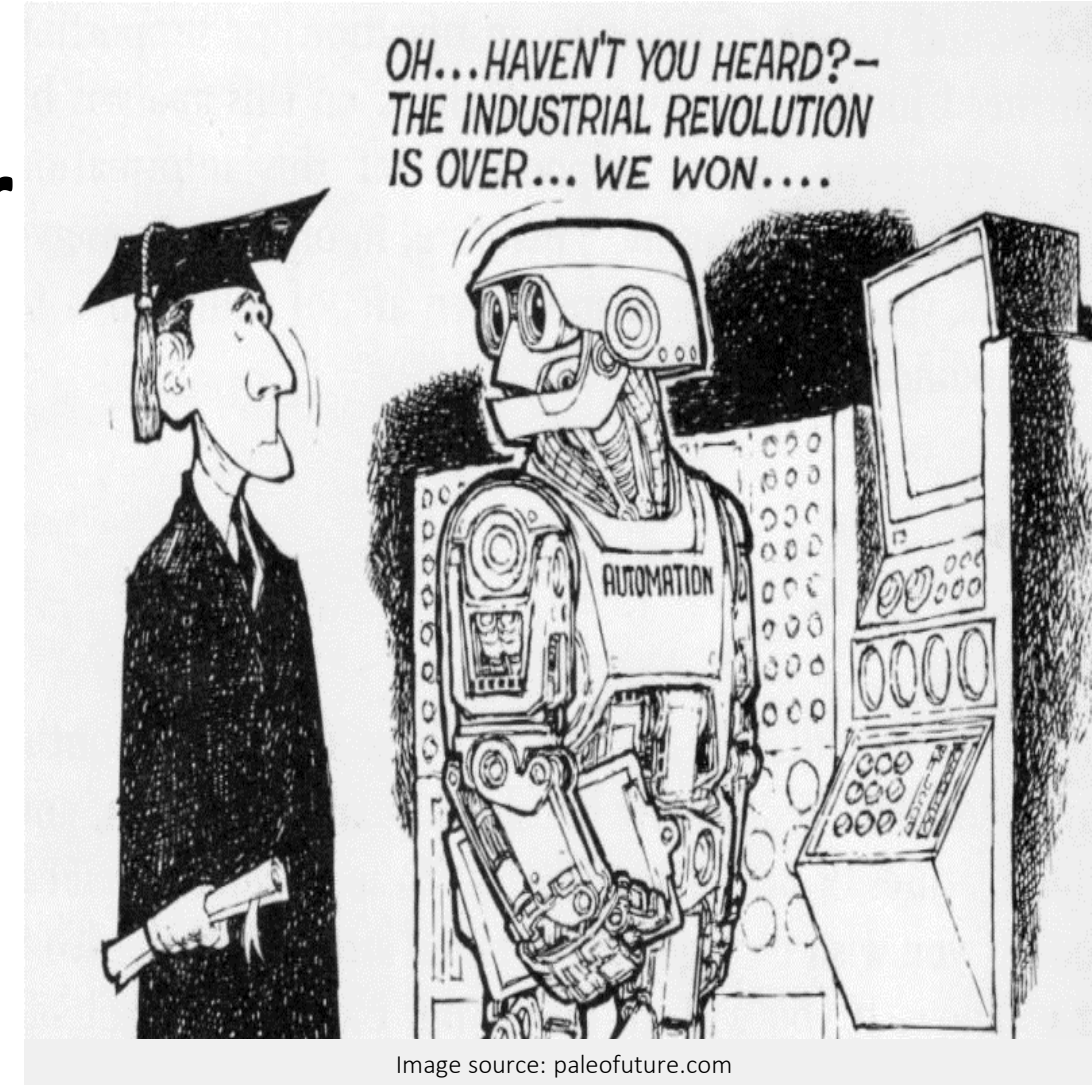
Supplementary Slides

- Doug Riecken, besides emergent behavior, also look at causal reasoning.
 - Why?
 - A baby (newborn) seems to be able to reason about certain things.
 - At some age, babies can do a lot of things that they were not specifically taught. Is that emergence?
- Ted
 - Comparison between ants and fish.
 - Hunger overcomes fear.
 - Learn how to inhibit fear when thirsty or hungry instead of programming it in. How can one encode it?
- Kevin Schmidt:
 - Recommends to read Randy O'Reily's paper on motivation and how it drives behaviors.
 - BU – from O'Reily's paper: also look at implementing goal-selection: is goal-selection similar to action-selection?

Background: What Is Autonomy?

Autonomy is the **delegation of agency from a user, either human or a software agent, to a software agent that utilizes a certain level of knowledge of the user's intents and desires.**

Similar to “commander’s intent” from military terminology.



A More Complex Critter

